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app: ① reduce the dimension of ~~matrix~~ vector
②

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Introduction to Funⁿ & Transformation

Functions: It maps input to output in a specific way.

Notation: A funⁿ f mapping elements from set X (domain) to set Y (codomain) is denoted by

$$f: X \rightarrow Y$$

e.g.,

$$\textcircled{a} f(x) = 2x + 3$$

$$\text{if } x = 5$$

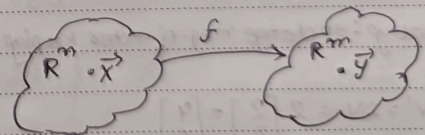
$$\therefore f(x) = y = 2 \times 5 + 3 = 13$$

$$\text{also, } 5 \in \mathbb{R}, 13 \in \mathbb{R}$$

$$\textcircled{b} f: \mathbb{R}^3 \rightarrow \mathbb{R}^2$$

$$f: \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3 \Rightarrow \begin{bmatrix} x+y \\ 6z \end{bmatrix} \in \mathbb{R}^2$$

Vector Transformations



$$f: X \mapsto Y$$

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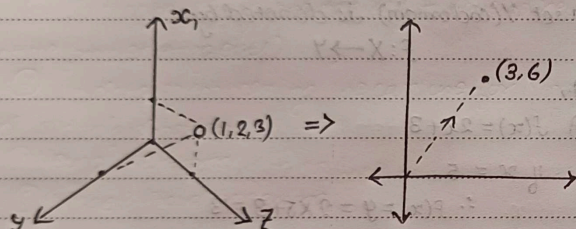
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e.g.,

$$f(x, y, z) = f(x+y, z, z)$$

$$\therefore f\left(\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}\right) = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$$



$$(1, 2, 3) \in \mathbb{R}^3 \Rightarrow (3, 6) \in \mathbb{R}^2$$

"This is called vector transformation"

$$T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$$

$\Rightarrow$ vector transformation refers to operations that map f from one space to another, often changing their magnitude, dirⁿ or both.

$\Rightarrow$ appⁿ: ① scaling: to change mag. of vector keeping dirⁿ same

$$V' = 2V = 2 \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 4 \\ 6 \end{bmatrix}$$

used in data normalization

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② Rotation: $\rightarrow$ image processing

$$V = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \in \mathbb{R}^2 \quad V' = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \Rightarrow 90^\circ \text{ rotation}$$

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Linear Transformation:

$\Rightarrow$ It is a f b/w two vector spaces that preserves the operations of vector addition and scalar multiplication. This means that if T is a linear transfⁿ from a vector space V to vector space W ; $T: V \rightarrow W$; then for any vectors it shall satisfy two properties given below:

① Additivity:

$$T(u+v) = T(u) + T(v)$$

② Homogeneity:

$$T(cv) = cT(v)$$

where,

$u, v \in V$ & c is a scalar value

$\Rightarrow$ e.g., ③ Reflection:

$$X = \begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2 \quad \therefore T(X) = \begin{bmatrix} -x \\ y \end{bmatrix}$$

Here, the transformation can be expressed as:

$$A = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \quad \therefore T(X) = AX \\ = \begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} -x \\ y \end{bmatrix}$$

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Why Linear Transformation?

⇒ Reasons: → it helps to maintain variance.

- ① To convert higher dimension (no. of features) to lower dimensions.
- ② To create feature engineering (to develop new feature from existing ones)
- ③ In data preprocessing we can use for Normalization and standardization.
- ④ Neural Networks
- ⑤ Image & signal processing

Vector Length & Vector Unit:

⇒ unit vector → vector of length 1.

⇒ if $\vec{x} \in \mathbb{R}^n$ $\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$

$$\therefore \|\vec{x}\| = \sqrt{x_1^2 + x_2^2 + \dots + x_n^2}$$

magnitude or vector length

⇒ if $\|\vec{u}\| = 1$ or $\hat{u}$

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⇒ To convert a vector of n dimension ($\vec{V}$) to a unit vector (say $\vec{u}$)

$$\therefore \text{if } \vec{V} = \begin{bmatrix} V_1 \\ V_2 \\ \vdots \\ V_n \end{bmatrix} \quad \therefore \vec{u} = \frac{1}{\|\vec{V}\|} \cdot \vec{V}$$

dot product

e.g., $\vec{V} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} \in \mathbb{R}^3$

$$\therefore \|\vec{V}\| = \sqrt{1^2 + 2^2 + 0} = \sqrt{5}$$

$$\therefore \vec{u} = \frac{1}{\sqrt{5}} \cdot \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{5} \\ 2/\sqrt{5} \\ 0 \end{bmatrix}$$

Use,

$$\|\vec{u}\| = \sqrt{(1/\sqrt{5})^2 + (2/\sqrt{5})^2 + 0} = 1 = \hat{u}$$

⇒ when to use?

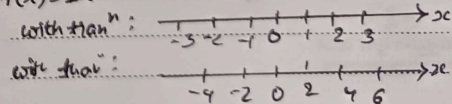
① Normalization → to improve optimization.

1. T. Visualization: [Note: ① origin must remain fixed
② all lines must remain line.]

⇒ if $T: \mathbb{R} \rightarrow \mathbb{R}$ (1 dimension)

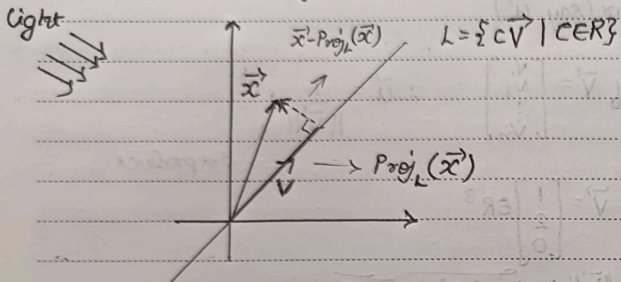
$$T(x) = 2x$$

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Introduction to Projections:



$\text{Proj}_L(\vec{x}) = \text{Proj of } \vec{x} \text{ on line } L$

where,

$$L = \{c\vec{v} \mid c \in \mathbb{R}\}$$

Also,

$\text{Proj}_L(\vec{x}) = \text{some vector in line where } \vec{x} - \text{Proj}_L(\vec{x})$

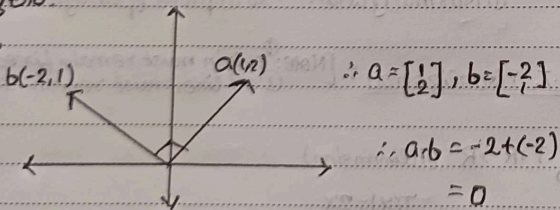
is perpendicular to L .

$$= (\vec{x} - c\vec{v}) \cdot \vec{v} = 0 \Rightarrow c = \frac{\vec{x} \cdot \vec{v}}{\vec{v} \cdot \vec{v}}$$

$\Rightarrow$ A vector perpendicular to another vector given dot product

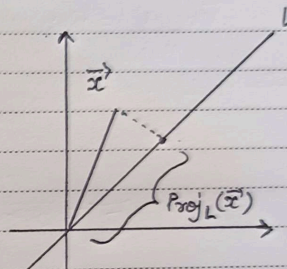
as zero.

E.g.,



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$$\Rightarrow \therefore \text{Proj}_L(\vec{x}) = c\vec{v} = \left(\frac{\vec{x} \cdot \vec{v}}{\vec{v} \cdot \vec{v}} \right) \cdot \vec{v}$$



E.g.,

$$\vec{v} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \vec{x} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

$$\therefore L = \left\{ c \begin{bmatrix} 2 \\ 1 \end{bmatrix}, c \in \mathbb{R} \right\}$$

$$\therefore \text{Proj}_L(\vec{x}) = \left(\frac{\begin{bmatrix} 2 \\ 3 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 1 \end{bmatrix}}{\begin{bmatrix} 2 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 1 \end{bmatrix}} \right) \cdot \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \frac{7}{5} \cdot \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 14/5 \\ 7/5 \end{bmatrix} = \begin{bmatrix} 2.8 \\ 1.4 \end{bmatrix}$$